MODULE 1

Population Growth Rate:

Population Growth

Formulas:

Rate	Population Growth	Exponential Growth	Logistic Growth
dY/dt	dN/dt = B – D	$\frac{dN}{dt} = r_{max} N$	$\frac{dN}{dt} = r_{max} N\left(\frac{K-N}{K}\right)$

dY = amount of change
t = time
B = birth rate
D = death rate

N = population size
K = carrying capacity
$\mathbf{r_{max}}$ = maximum per capita growth rate of population

$$\frac{dN}{dt} = \frac{\Delta N}{\Delta t} = \frac{change\ in\ population\ size}{change\ in\ time} = \textbf{population growth rate}$$

1. Exponential Growth

Exponential growth occurs when a population increases at a rate proportional to its current size. This is common in populations with unlimited resources.

Equation:

$$N(t) = N_0 e^{rt}$$

- $N(t)$: Population at time t
- N_0: Initial population
- r: Growth rate
- t: Time

Question 1:

Population growth rate **dN/dt** is the *change in the population size* (**N**) over some *time* (**t**) interval.

Population Growth Rate (dN/dt) = birth rate (B) – death rate (D)

Birth rate (**B**) is the proportion of individuals born in a population (*over a period of time*).
Example: if there are 5 births among 10 individuals,

$$B = 5/10 = \boxed{\mathbf{0.5}}$$

Death rate (D) is the proportion of individuals dying in a population (*over a period of time*).
Example: if 4 of 10 individuals die,

$$D = 4/10 = \boxed{\mathbf{0.4}}$$

$$\begin{aligned} \text{Thus,}\quad \mathbf{dN/dt} &= \mathbf{B - D} \\ \mathbf{dN/dt} &= \mathbf{0.5 - 0.4} \\ \mathbf{dN/dt} &= \boxed{\mathbf{0.1}} \end{aligned}$$

Change in population can be calculated by multiplying the growth rate (**dN/dt**) by the original population size (**N**)

Change in Population = (**dN/dt**) **N**

In this example, change in population = (**dN/dt**) **N** = **0.1(10)** = 1,
so the population has increased by one individual in that time period.

To determine the *size of the population at the end of the time period*,
add the population size (**N**) to the change in the population

$$\begin{aligned} &= \mathbf{N + (dN/dt)\ N} \\ &= 10 + (0.1)10 \\ &= 10 + 1 \\ &= \boxed{\mathbf{11}} \end{aligned}$$

Question 2

A population of bacteria is growing exponentially with a growth rate r = 0.15 per hour. If the population size at time t = 0 is 1,000, calculate the rate of population growth at t = 2 hours

Solution:

First, calculate the population size at $t = 2$ hours:

$$N(t) = N_0 e^{rt}$$

$$N(2) = 1,000 \cdot e^{0.15 \cdot 2}$$

$$N(2) = 1,000 \cdot e^{0.3}$$

$$N(2) = 1,000 \cdot 1.3499$$

$$N(2) \approx 1,350$$

Question 3

In a population of **750** fish, **25** die on a particular day while **12** were born.

a) What is the *death rate* **(D)** for the day? *Round your answer to the nearest hundredth.*

$$\frac{25}{750} = 0.033 = \underline{0.03}$$

b) What is the *birth rate* **(B)** for the day? *Round your answer to the nearest hundredth.*

$$\frac{12}{750} = 0.016 = \underline{0.02}$$

c) What is the *growth rate* of the population (B – D)? *Round your answer to the nearest hundredth.*

$$\frac{dN}{dt} = B - D$$

$$= 0.02 - 0.03 = \underline{-0.01}$$

Question 4

A population of deer is growing logistically with a carrying capacity K = 1,000, a growth rate r = 0.1 per year, and a current population size N = 500. Calculate the rate of population growth

Solution:

Use the logistic growth formula:

$$\frac{dN}{dt} = r\left(\frac{K-N}{K}\right)N$$

- $r = 0.1$,
- $K = 1,000$,
- $N = 500$.

$$\frac{dN}{dt} = 0.1\left(\frac{1,000-500}{1,000}\right)500$$

$$\frac{dN}{dt} = 0.1\left(\frac{500}{1,000}\right)500$$

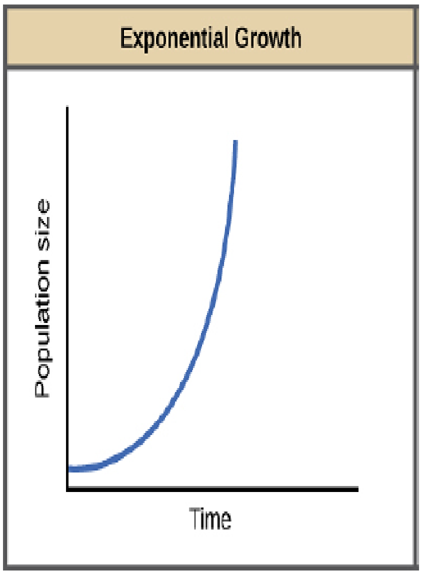
$$\frac{dN}{dt} = 0.1 \cdot 0.5 \cdot 500$$

$$\frac{dN}{dt} = 25 \text{ individuals per year}$$

The population is growing at a rate of **25 individuals per year**.

Growth types

Exponential growth is continuous population growth in an environment where resources are unlimited; it is **density-independent growth**.

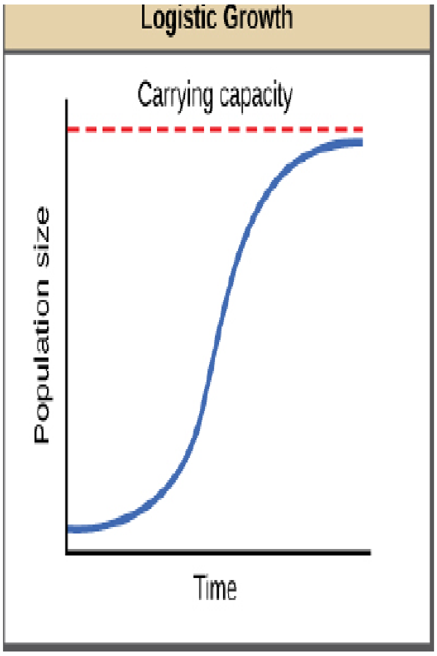


Most density-independent factors are **abiotic**, or nonliving, and include:

temperature	DO - dissolved O_2
natural disasters	pollution

Formula: $$\frac{dN}{dt} = r_{max} N$$

Logistic growth is continuous population growth in an environment where resources are limited; it is **density-dependent growth.**



Most density-dependent factors (*a limiting factor that depends on population size*) are mainly **biotic**, or living, and include:

disease	competition
predation	migration

Formula: $$\frac{dN}{dt} = r_{max} N\left(\frac{K-N}{K}\right)$$

Question 5

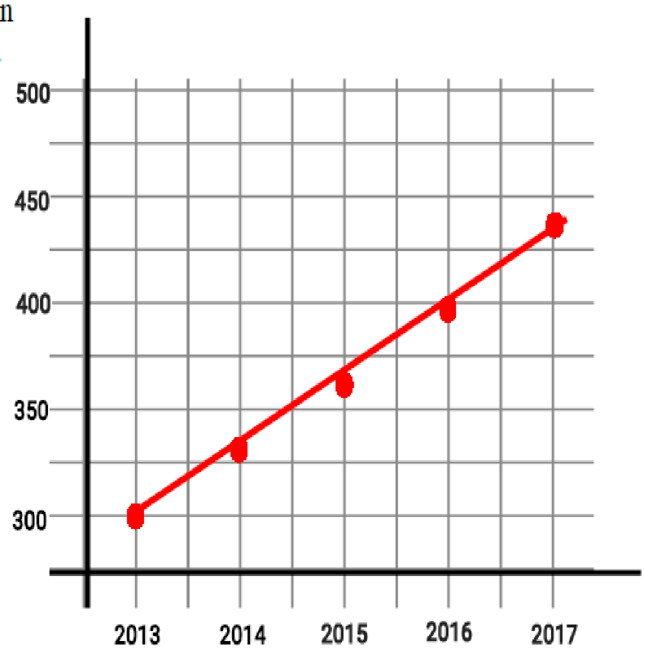
There are **300** falcons living in a certain forest at the beginning of 2013. The population is under carrying capacity. If the maximum per capita growth rate (r_{max}) = 0.1 falcons/year, predict the population size of the falcon population each year for the next four years.

$$\frac{dN}{dt} = r_{max} N$$

2014	2015	2016	2017
300 x 0.1 = 30 300 + 30 = 330	330 x 0.1 = 33 330 + 33 = 363	363 x 0.1 = 36 363 + 36 = 399	399 x 0.1 = 40 399 + 40 = 439

(a) Using the information from above, fill in the table below and construct the graph.

Year	Population Size
2013	300
2014	330
2015	363
2016	399
2017	439



(b) Find the **average rate of change** *(slope)* for the falcon population from 2013 to 2018.

Round your answer to the nearest tenth.

$$\text{Slope} = m = \frac{rise}{run} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{439 - 300}{2017 - 2013} = \frac{139}{4} = \underline{34.8}$$

QUESTION 6.

A population of bacteria starts with 100 individuals and grows at a rate of 10% per hour. Calculate the population after 5 hours.

Solution:

$$N(t) = 100 \cdot e^{0.10 \cdot 5}$$

$$N(t) = 100 \cdot e^{0.5}$$

$$N(t) = 100 \cdot 1.6487$$

$$N(t) \approx 165$$

Numerical

A population of bacteria is growing exponentially with a growth rate r = 0.2 per hour. If the current population size is 500, calculate the rate of population growth

Solution:

$$\frac{dN}{dt} = rN$$

$$\frac{dN}{dt} = 0.2 \cdot 500$$

$$\frac{dN}{dt} = 100 \, \text{individuals per hour}$$

The population is growing at a rate of **100 individuals per hour**.

Module 2:

Atom Economy

1. Calculate the atom economy for the following reactions. The desired products in the two reactions are shown in red.

:u.

a) $C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$

Solution:

$$\text{Atom Economy} = \frac{\text{Molecular weight of desired product}}{\text{Molecular weight of all reactants}} \times 100$$

$$= \frac{123}{78+63} \times 100$$

$$= 87.2\ \%$$

b) $CH_3COCl + C_2H_5NH_2 \rightarrow CH_3CONHC_2H_5 + HCl$

Solution:

$$\text{Atom Economy} = \frac{\text{Molecular weight of desired product}}{\text{Molecular weight of all reactants}} \times 100$$

$$= \frac{87}{78.5+45} \times 100$$

$$= 70.4\ \%$$

PROBLEM 2.

Table 2 shows the calculation of the overall atom economy; it is 77% - almost double that of the Boots' synthesis.

Reagent		**Used in ibuprofen**		**Unused in ibuprofen**	
Formula	**M_r**	**Formula**	**M_r**	**Formula**	**M_r**
$C_{10}H_{14}$	134	$C_{10}H_{13}$	133	H	1
$C_4H_6O_3$	102	C_2H_3O	43	$C_2H_3O_2$	59
H_2	2	H_2	2		0
CO	28	CO	0		0
Total		**Ibuprofen**		**Waste products**	
$C_{15}H_{22}O_4$	266	$C_{13}H_{18}O_2$	206	$C_2H_4O_2$	60

Table 2 Atom economy in the green synthesis of ibuprofen

$$\text{Atom Economy} = \frac{\text{Molecular weight of desired product}}{\text{Molecular weight of all reactants}} \times 100$$

$= (206/266) \times 100 = 77\ \%$

3. Calculate the percentage atom economy for the following reaction with respect to cinnamaldehyde.

$C_6H_5CHO + CH_3CHO \rightarrow \mathbf{C_6H_5CH{=}CHCHO} + H_2O$

(Ans: 88 %)

Module 4-- Margin of Safety(4.4)

Margin of Safety

1. A drug is effective for 99% of people at 12 mg and toxic to 1% of people at 10 mg. Determine the margin of safety of the drug.

Solution:

$MOS = LD_1 / ED_{99} = 10/ 12 = 0.83$

2. A new sedative has lethal dose of 150 mg/Kg and effective dose of 25mg/Kg. Determine the margin of safety of this sedative.

Solution:

$MOS = LD_1 / ED_{99} = 150/ 25 = 6$

3. Which of these drugs is safer?

   Drug A : $LD_1 = 50$; $ED_{99} = 10$

   Drug B: $LD_1 = 100$; $ED_{99} = 40$

   Solution:

   Drug A, $MOS = LD_1 / ED_{99} = 50 / 10 = 5$

   Drug B, $MOS = LD_1 / ED_{99} = 100 / 40 = 2.5$

   Drug A that has MOS greater than Drug B is a much safer drug.

Module 5:
TYPE 1-Solar Cell Efficiency

1. A single solar cell (10 cm × 10 cm) produces a voltage of 0.5 V and a current up to 2.5 A. If the solar insolation is 800 W/m^2 , calculate the efficiency of the solar cell is.

Solution:

Efficiency, η = (P_out/ P_in) * 100

Area: 0.1m x 0.1m = 0.01 m^2.

Power Output (P_out): 0.5V * 2.5A = 1.25 W. (maximum current *maximum voltage)

Solar Input (P_in): 0.01 m^2 * 800 W/m^2 = 8 W. (Panel area*Solar insolation)

η = (P_out/ P_in) * 100
= (1.25 W / 8 W) * 100 = 15.625%

Type 2-Area calculation

Type II – Calculate Area

2. A single solar cell on illumination by insolation of about 800 Wm^{-2} produces a voltage of 0.5 V and a current up to 2.0 A. The efficiency of the solar cell is 12.5% . The area of the cell is:

A. 2×10^{-2} m^2

B. 5×10^{-3} m^2

C. 4×10^{-4} m^2

D. (4) 10^{-2} m^2

Answer: (D) 10^{-2} m^2

Explanation:

Given data's.

SI = 800 w/m^2

Voltage (V) = 0.5 V

Current (I) = 2.0 A

N = 12.5%

Area = ?

$\eta = \frac{P_{out}}{P_{in}}$ x 100 ... (i)

Solar Insulation (S. I) = $\frac{P_{in}}{Area}$ ... (ii)

P_{out} = Voltage (V) x Current (I) ... (iii)

Using (i), (ii) and (iii) we can deduce.

$$\eta = \frac{V*I}{S.I*Area}$$

$$\frac{12.5}{100} = \frac{0.5*2}{800*A}$$

$$A*12.5*800 = 0.5*2*100$$

$$A = \frac{100}{10000}$$

$$A = 10^{-2}\ m^2$$

The Area of the cell is $10^{-2}\ m^2$ Hence option (D) $10^{-2}\ m^2$ is the correct answer.

TYPE III-CALCULATE POWER OUTPUT

3.Solar insolation on a rectangular module (1.5 m x 2.0 m) of photovoltaic cells is 550 W/m^2. If the efficiency of the cells is 12% What is the power output of the module?

Solution:

Efficiency Formula Rearranged: P_out = (η / 100) * P_in

Solar Input (P_in): 3 m^2 * 550 W/m^2 = 1650 W.

Power Output (P_out): (12 / 100) * 1650 W = 198 W

4. A 1m^2 panel has 20% efficiency at 1000 W/m^2 irradiance. What's its maximum power (P_{out})?

Solution:

Efficiency Formula Rearranged: P_out = (η / 100) * P_in

Solar Input (P_in): 1 m^2 * 1000 W/m^2 = 1000 W.

Power Output (P_out): (20 / 100) * 1000 W = 200 W

USING STANDARD CONDITIONS:ASSUME-Standard Test Conditions is 1000 W/m^2

5. A solar panel has a maximum power output of 400 W and measures 2 square meters. If the solar irradiance under Standard Test Conditions is 1000 W/m^2, what is the efficiency of the solar panel?

(Ans: Efficiency = 20%)

6. You need a photovoltaic system to generate a total of 5 kW of power. If you are using solar panels that are 17% efficient, how much land area is required for the installation under solar radiation?

(Ans: Area = approx 29.41 sq. m.)